Learning to Generate Realistic Hyperspectral Noise for Denoising Enhancement

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***Abstract*—Hyperspectral image (HSI) denoising in real-world scenarios is a significant concern in the field of remote sensing. Recently, deep-learning (DL)-based HSI denoising methods have provided an effective solution. However, the scarcity of real "clean–noisy" data remains a major challenge. Typically, most methods rely on synthetic datasets, which are of low cost to generate but low quality for training, resulting in denoisers lacking generalization capabilities on real HSIs. On the contrary, capturing real HSIs yields data of high quality but also high cost. In this article, we aim to discover a "low-cost, high-quality" method to efficiently generate realistic noisy HSIs. We propose an adversarial denoiser-aided noise generation network (ADNG-Net) to leverage the joint distribution of clean–noisy image pairs to guide noise generation. Specifically, we approximate the joint distribution through two different factorized forms, formulated as a generator mapping the clean images to the corresponding noisy ones and a denoiser mapping the noisy images to their clean counterparts, which are obtained through adversarial training. For better noise generation, we also develop a noise generation network that takes the HSI imaging mechanism into consideration. In addition, we design customized loss functions to enhance training. Extensive experiments have been conducted to demonstrate the effectiveness of the proposed method. Notably, the noise generated by our method is highly similar to the real HSI noise in terms of both distributional characteristics and visual effects, and the generated "clean–noisy" pairs can significantly enhance the performance of HSI denoisers on various datasets.**

***Index Terms*—Deep learning (DL), generative adversarial network (GAN), hyperspectral image (HSI) denoising, noise generation, signal-dependent (SD) noise.**

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I. Introduction

HYPERSPECTRAL images (HSIs) typically have tens to hundreds of bands, providing richer and more detailed spectral information for each object, compared to RGB and multispectral images. Thus, HSIs are conducive to the distinction and recognition of objects, and have been widely applied in various domains, such as remote sensing [1], vegetation detection [2], and classification [3]. However, due to the limited photon counts for each band and the complex imaging process, HSIs are easily degraded by various noises. This degradation presents difficulties for the aforementioned downstream tasks. Therefore, HSI denoising, as an image preprocessing step, is crucial for subsequent data analysis and interpretation.

Most HSI denoising methods can be summarized within the Bayesian framework. Assume that the image to be denoised is $\mathbf{Y}$ and the underlying clean image is $\mathbf{X}$. The purpose of denoising is to recover $\mathbf{X}$ from $\mathbf{Y}$, that is, to estimate the conditional distribution $p(\mathbf{X}|\mathbf{Y})$. According to Bayes' formula, $p(\mathbf{X}|\mathbf{Y}) \propto p(\mathbf{Y}|\mathbf{X})p(\mathbf{X})$, where $p(\mathbf{Y}|\mathbf{X})$ is the likelihood term and $p(\mathbf{X})$ is the prior term. Thus, the traditional HSI denoising methods are mainly developed from these two terms. One kind of method focuses on modeling the likelihood term, such as NMoG [4] and DP-GMM [5], which models the mechanism of noise generation by assuming reasonable distributions. The other kind of methods focuses on encoding the prior characteristics of HSIs, such as local smoothness [6], [7], [8], low-rankness [9], [10], [11], sparsity [12], [13], and spectral correlation [8], [14]. However, these two methodologies are constrained by subjective assumptions regarding the noise mechanism and image priors, which may deviate from real-world scenarios.

In recent years, a large number of supervised deep-learning (DL)-based methods have shown surprising results in the HSI denoising task. By making full use of the correspondence between the noisy images and clean counterparts, these methods are designed to implicitly depict priors and directly learn to estimate $p(\mathbf{X}|\mathbf{Y})$ by deep neural networks. Apparently, the performance of a deep denoiser depends heavily on the number and the quality of clean–noisy HSI pairs for training. However, for HSIs, it is very difficult to obtain such high-quality paired training data. Specifically, researchers have to adjust equipment parameters to obtain the clean image and the noisy image in the same scene, and try to avoid problems,

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like camera shake and brightness deviation. Besides, such a capturing process is generally of high cost.

Due to the difficulties of obtaining high-quality paired data, currently, almost all supervised denoising methods artificially synthesize the noisy HSIs based on the clean ones to obtain paired training data. The majority of existing research utilizes additive noise models, such as independent and identically distributed (i.i.d.) Gaussian noise, non-i.i.d. Gaussian noise, stripe noise, impulse noise, and mixed noise. Such a strategy of introducing noise is significantly more cost-effective compared to the real-scene noise acquisition process. However, although these synthetic noises resemble the complex real HSI noise to a certain extent, they still exhibit notable differences from the real ones [15]. For example, the synthetic additive noise is signal-independent (SI), which ignores the potential signal-dependent (SD) components predominated under some hyperspectral scanners. As a result, the trained denoiser exhibits limited generalization ability.

Based on the aforementioned discussion, capturing real HSIs yields data of high quality but high cost, while synthetic datasets are of low cost but also relatively low quality. Hence, our research aims to develop a "low-cost and high-quality" approach to efficiently generate realistic HSIs. Inspired by DANet [16], we adopt a deep generative model to estimate the conditional noise distribution, $p(\mathbf{Y}|\mathbf{X})$, and train it within a dual adversarial training framework. Specifically, we factorize the joint distribution $p(\mathbf{X}, \mathbf{Y})$ by two directions of noise removal and noise generation, i.e., $p(\mathbf{X}|\mathbf{Y})p(\mathbf{Y})$ and $p(\mathbf{Y}|\mathbf{X})p(\mathbf{X})$, which are modeled by a denoiser and a noise generator, respectively. Then, we jointly train the denoiser and the generator to approximate the joint distribution $p(\mathbf{X}, \mathbf{Y})$ based on the two factorizations, in a dual adversarial fashion. Different from DANet, we pay more attention to the special characteristics of HSI noise and develop an effective generator regarding these characteristics, as well as loss functions for better training. In specific, we develop a transformer-based HSI noise generator with band-dependent random input, such that the spatial correlations, spectral differences, signal-dependency, and local randomness of the HSI noise are expected to be well characterized. Furthermore, for a more efficient training, we introduce a GAN loss and a Kullback–Leibler (KL) divergence loss to constrain the statistical characteristics of the generated noise by the generator.

The learned generator enables the random generation of realistic noisy images based on any given clean HSI, serving as a generalized tool for data augmentation for real noise removal. The new clean–noisy image pairs with the generated noisy HSIs can further enhance the denoising network, and also provide a better training set for real-world HSI denoising compared with traditional additive synthesis noise.

The main contributions of this work are summarized as follows.

1) We adopt a dual adversarial framework to train an HSI noise generator that fully utilizes the information of data pairs, with the aid of an HSI denoiser. Within the framework, the generator can generate noisy HSIs whose distribution is close to that of the real ones, without the traditional manual assumption of noise distribution. To our best knowledge, this is the first work employing a DL-based generative method for HSI noise generation.
2) For better generating the HSI noise, we design a transformer-based generator with band-dependent random input, such that the special characteristics of HSI noise can be well described. In addition, we also introduce a GAN loss and a KL divergence loss to constrain the statistical structures of the generated noise for efficient training.
3) Our proposed noise generation method produces realistic noisy HSIs for the HSI denoising task. Experiments show that the generated noisy HSIs are visually and quantitatively similar to real ones, significantly improving the performances of DL-based denoisers across various real HSI denoising datasets.

II. Related Works

Since our research focuses on noise generation for HSI denoising tasks, we review research works from three key aspects relevant to this topic: HSI noise removal, HSI noise estimation, and image noise generation.

A. HSI Noise Removal

HSI denoising methods can be broadly categorized into model-driven and data-driven approaches. The model-driven approaches are mainly based on priors about the underlying images as well as noise modeling. For image priors, typical examples include spatial–spectral total variation (TV) [7], [8], nonlocal similarity [17], sparse representation [6], [12], [18], and low-rank (LR) prior [9], [10], [11], [19], [20]. For example, Yuan et al. [7] proposed a denoising algorithm employing a spectral–spatial adaptive TV model, considering both the spectral noise differences and spatial information differences; Qian and Ye [18] incorporated both the nonlocal similarity and spectral–spatial structure of HSI into sparse representation; Wang et al. [6] proposed the TV-regularized LR tensor decomposition (LRTDTV) method, which adopted tensor Tucker decomposition to capture the global spectral correlation and characterize the piecewise smooth structure; and He et al. [20] designed a nonlocal meeting global (NGmeet) model, integrating the spatial nonlocal similarity and global spectral LR prior.

As for noise modeling, early methods primarily employed an L2 loss term to constrain the deviation between the reconstruction and the original HSI [21], [22], [23], which implies an i.i.d. Gaussian assumption for noise from a statistical perspective. The L1 loss term was also generally used to enhance robustness, assuming an i.i.d. Laplacian noise distribution [24], [25], which works well for sparse noise, such as stripes and deadlines. To handle more complex noise cases, a mixture of Gaussians (MoGs) was employed to describe noise. For example, Chen et al. [4] proposed to model the noise as a mixture of non-i.i.d. Gaussians, and Yue et al. [5] modeled the HSI noise structure through a two-level hierarchical Dirichlet process (HDP) to further consider the various complexities of MoGs in different HSI bands. In general, the above methods primarily employed iterative optimization techniques to solve

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the denoising problem, without relying on paired data for supervised training.

Recently, data-driven methods utilizing DL have emerged as promising approaches to HSI denoising. Unlike model-driven methods, these approaches do not rely on handcrafted priors. Instead, they learn the deep prior from data and establish a mapping from the noisy HSIs to the clean ones through end-to-end training. Along this line of research, Chang et al. [26] first employed DL for HSI denoising by simply using a 2-D convolutional neural network (CNN). Then, Yuan et al. [27] proposed HSI-DCNN, which performed denoising band by band, utilizing the current band picture along with the adjacent bands to extract the spatial and spectral correlation. Such a denoising framework has been inherently used in later methods [28], [29], [30]. Rather than directly processing the original HSIs, Lin et al. [31] converted the HSIs to low-dimensional representations before applying CNNs on the reduced images. T3SC [32] and MAC-Net [33] transformed the optimization of model-driven methods into neural architectures, encoding domain knowledge with image priors. In the case of 3D-ADNet [30], the spatial–spectral features were extracted by combining 3-D convolution and self-attention. QRNN3D [34] developed a quasi-recurrent pooling function together with 3-D convolution to capture both the structural spatial–spectral correlation and the global spectral correlation. Beyond the CNNs, recent research studies have also introduced the transformer architectures to HSI denoising. Wang et al. [35] first employed the transformer structure based on the Swin transformer [36] for HSI restoration. TRQ3DNet [37] combined the QRU [34] block and Uformer [38] to further explore global spatial correlation. Hider [39] and HSDT [40] performed 3-D self-attention to enrich the model with powerful capabilities. As discussed before, though these DL-based methods have shown promising performance on the HSI denoising task, they are sensitive to the quality of training data.

B. Noise Estimation

Noise estimation methods typically involve defining a parametric model for random noise and estimating its parameters from noise realizations, which are generally treated as an optional preprocessing step for traditional HSI denoising methods. In early methods, noise was assumed to be SI [41] and spatially stationary with zero mean. The model parameters were simply the noise variances or covariances across bands [41], [42], [43], which were estimated mainly according to the homogeneous area (HA) methods [44]. Gao [45] first employed a box counting procedure to estimate the noise variance for each band by considering homogeneous regions. Roger and Arnold [46] extended this approach by selecting the homogeneous blocks in an automatic way and applied the multiple linear regression (MLR) to obtain the estimated noise. Mahmood et al. [47], [48] further modified MLR by including noise covariance in the analysis. Gao et al. [49] utilized an object-seeking algorithm [50] and Aiazzi et al. [51] modified the scatter plot method [52] to identify homogeneous regions. Other methods estimated the noise parameters by operating in the spectral domain to separate noise and signal, assuming the smoothness of image texture, e.g., spectral decomposition approaches [53], [54] and data-masking approaches [55]. These research studies made restricted assumptions about the HSI noise and neglected the SD component of it. However, subsequent researches showed that, with the update of hyperspectral sensors, SI noise can be suppressed well [56], [57]. Thus, photon noise, which depends on the useful signal level, is equally important to consider.

The signal-dependency of digital image noise has been studied for a considerable period of time [58]. For HSI noise, Alparone et al. [56] extended the two-parameter SD noise model [58] to characterize the noise in each band of HSIs. This study, along with [59], modified the scatter plot technique to estimate the SI and SD noise parameters. Uss et al. [60] estimated both SI and SD band noise components by linear fit, using local noise variances obtained based on the Brownian motion model. A similar but more convenient method was proposed in [61] to estimate the SI and SD parameters by directly applying maximum-likelihood (ML) to the estimated noise extracted by MLR, and thus avoids selecting HAs. Similar techniques such as MLR, HA, and ML was also adopted in [62], [63], and [64].

Theoretically, the aforementioned HSI noise estimation methods can also be used to generate noise, provided the noise model can fit the real noise well. However, due to the less flexibility of preassumed parametric models, such a strategy encounters difficulties in accurately depicting the distribution of the real HSI noise.

C. Noise Generation

Noise generation plays an important role in augmenting real datasets and enhancing the performance of denoising techniques. In the domain of color images, ongoing research mainly focuses on physics-based and statistics-based methods. Physics-based methods model the noise formation process to simulate different components of noise [65], [66], [67], [68]. For example, Wei et al. [66] presented an accurate noise formation model based on the characteristics of CMOS photosensors and calibrated the noise parameters for modern digital cameras. Another research study [67] developed the pattern-aligned patch sampling to handle the SI noise and achieved high-bit reconstruction. Statistics-based methods primarily use deep probabilistic models to capture the complex distribution of noise. For example, DANet [16] learned the mapping of the denoiser and the generator within a dual adversarial framework and augmented the training dataset to improve the performance of the denoiser. Noise Flow [69] learned the shift of distribution between the synthetic and the desired noise by utilizing a normalizing flow network. Recent work [70] employed a diffusion model to synthesize realistic noise under different camera settings.

For HSI noise generation, a recent research study [15] physically formulated a four-parameter-based noise model considering multiple stages of HSI noise formation. However, this approach poses operational challenges and has limitations regarding the capability of the noise model. To the best of our knowledge, there are currently no statistics-based methods for HSI noise generation due to its complex characteristics and the

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lack of sufficient real-world paired datasets. Therefore, we aim at addressing this issue by building a deep generative model from a statistical modeling perspective, inspired by the related studies for color images.

III. Methodology

In this section, we present our method for generating realistic HSI noise. Specifically, we first factorize the joint distribution of the "clean–noisy" HSI pairs from both the noise generation and the noise removal perspectives, which can be approximated by virtue of a deep generator and a deep denoiser, respectively. Then, we introduce the dual adversarial framework for simultaneously training the generator and the denoiser, especially the loss terms for the generator, which specifically consider the properties of HSI noise. Finally, we discuss the network architectures, and mainly focus on designing the HSI noise generator that is expected to well characterize the real HSI noise.

A. Formulation

In this part, inspired by the work of DANet [16] for color images, we factorize the joint distribution of the "clean–noisy" HSI pairs from two perspectives, i.e., the noise generation process $\mathbf{X} \Rightarrow \mathbf{Y}$, and the noise removal process $\mathbf{Y} \Rightarrow \mathbf{X}$, respectively, which is the base of the noise generator training as will be discussed in Section III-B. Here, we denote the clean and the noisy HSIs as $\mathbf{X} \in \mathbb{R}^{B\times H\times W}$ and $\mathbf{Y} \in \mathbb{R}^{B\times H\times W}$, where B, H, and W represent the band counts, height, and width of the HSI.

1) Noise Generation Perspective: Since noise generation is a "one-to-many" process, where one clean image can correspond to several noisy images, we introduce an extra hidden variable $\mathbf{Z}$ to the input of the generator to make the generation process random. This hidden variable also covers the random noise generated during the imaging process of hyperspectral cameras. Hence, the process of generating noise from $\mathbf{X}$ to $\mathbf{Y}$ can be viewed as the inference process of the conditional distribution $p(\mathbf{Y}|\mathbf{X},\mathbf{Z})$. Such a conditional distribution can be approximated by a deep generator G. Specifically, when a clean image $\mathbf{X}$ and a latent variable $\mathbf{Z}$ are provided as input, the output of the generator $G(\mathbf{X},\mathbf{Z})$ can be considered as a sample drawn from the approximated conditional distribution, denoted as $p_G(\mathbf{Y}|\mathbf{X},\mathbf{Z})$.

For RGB images, the acquisition methods of the three channels involve similar optical and ISP process, so we can simply assume that $\mathbf{Z}$ is independent of $\mathbf{X}$, and is directly drawn from some distribution $p(\mathbf{Z})$, which is what has been done in [16]. However, differing from RGB images, HSIs acquire rich 3-D information. Consequently, due to the varying absorption and reflection of light at different wavelengths, the number of photons obtained at different bands varies greatly, leading to different shot noise intensities across these bands. Therefore, it would also be more reasonable to assume that $\mathbf{Z}$ is dependent on $\mathbf{X}$, such that the different noise structures along the spectral bands can be described. According to the above analysis, we formulate the generation process of the noisy HSI as follows:

$$\mathbf{X} \sim p(\mathbf{X}), \quad \mathbf{Z} \sim p(\mathbf{Z}|\mathbf{X}), \ \hat{\mathbf{Y}} = G(\mathbf{X},\mathbf{Z}) \tag{1}$$

where $\hat{\mathbf{Y}} \in \mathbb{R}^{B\times H\times W}$ denotes the noisy HSI generated by G. Then, the joint distribution $p_G(\mathbf{X},\mathbf{Y})$ is obtained by marginalizing the latent variable $\mathbf{Z}$

$$\begin{aligned} p_G(\mathbf{X},\mathbf{Y}) &= \int_{\mathbf{Z}} p_G(\mathbf{Y} \mid \mathbf{X},\mathbf{Z}) p(\mathbf{Z} \mid \mathbf{X}) p(\mathbf{X}) \mathrm{d}\mathbf{Z} \\ &\approx \frac{1}{L}\sum_{i}^{L} p_G\left(\mathbf{Y} \mid \mathbf{X},\mathbf{Z}^i\right) p(\mathbf{X}) \end{aligned} \tag{2}$$

where $\mathbf{Z}^i \sim p(\mathbf{Z}|\mathbf{X})(i = 1, 2, \ldots, L)$, and the second line is due to the Monte Carlo approximation. In practice, L can even be set to 1 while still retaining a reasonable approximation to the joint distribution $p_G(\mathbf{X},\mathbf{Y})$.

2) Noise Removal Perspective: Similarly, the denoising process can be regarded as the inference process of the conditional distribution $p(\mathbf{X}|\mathbf{Y})$. Consequently, following [16], the factorization of the joint distribution from the noise removal perspective can be formulated as

$$p_R(\mathbf{X},\mathbf{Y}) = p_R(\mathbf{X}|\mathbf{Y})p(\mathbf{Y}) \tag{3}$$

where R represents the denoiser. For convenience, we will denote $\hat{\mathbf{X}} \in \mathbb{R}^{B\times H\times W}$ as the denoised HSI output by R in the following.

B. Adversarial Denoiser-Aided Noise Generation Network

In Section III-A, we have modeled the joint distribution of "clean–noisy" HSI pair, i.e., $p(\mathbf{X},\mathbf{Y})$, from the perspectives of both noise generation and noise removal. Now, our objective is to push $p_R(\mathbf{X},\mathbf{Y})$ and $p_G(\mathbf{Y},\mathbf{X})$ toward $p(\mathbf{X},\mathbf{Y})$, such that the noise generator G can be well trained with the aid of the denoiser D. To achieve this, we follow the dual adversarial training strategy introduced in [16], and specifically design loss functions to constrain the distribution of the generated noise. We call the proposed framework as adversarial denoiser-aided noise generation network (ADNG-Net), and depict its overall working mechanism in Fig. 1. Specifically, the proposed ADNG-Net is composed of four subnetworks, i.e., a generator, a denoiser, a joint discriminator, and a noise discriminator. For the generator's input, the hidden variable $\mathbf{Z}$ is synthesized based on spectral knowledge of the clean image $\mathbf{X}$ (see Section III-C2 for more details). Next, $\mathbf{X}$ and $\mathbf{Z}$ are concatenated along the spectrum dimension and fed as input to the generator G. Then, noisy output $\hat{\mathbf{Y}}$ is concatenated with $\mathbf{X}$ as the image pair $(\mathbf{X},\hat{\mathbf{Y}})$. Similarly, the denoised output $\hat{\mathbf{X}}$ by the denoiser R is concatenated with $\mathbf{Y}$ as the image pair $(\hat{\mathbf{X}},\mathbf{Y})$. Then, the generator G and denoiser R are simultaneously trained adversarially with the joint discriminator D_J, such that the joint distributions of both $(\hat{\mathbf{X}},\mathbf{Y})$ and $(\mathbf{X},\hat{\mathbf{Y}})$ are close to the true joint distribution $p(\mathbf{X},\mathbf{Y})$. In addition, the noise discriminator D_N also acts as an opponent to the generator G, to force the generated noise to be close to the true HSI noise in distribution.

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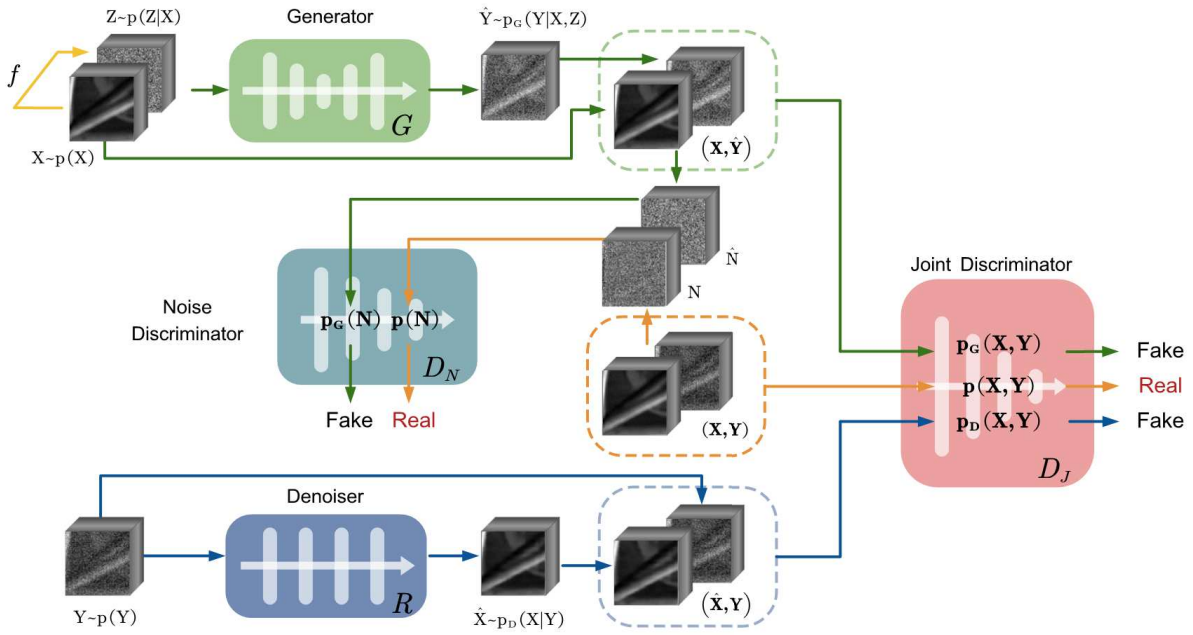


Fig. 1. Flowchart of our proposed ADNG-Net. It is composed of four subnetworks: a generator, a denoiser, a joint discriminator, and a noise discriminator.

1) Joint Distribution Approximation: The joint discriminator D_J tries to distinguish the real "clean–noisy" image pairs $(\mathbf{X}, \mathbf{Y})$ from the pseudo ones $(\hat{\mathbf{X}}, \mathbf{Y})$ and $(\mathbf{X}, \hat{\mathbf{Y}})$. On the contrary, the generator G and the denoiser R aim to produce outputs that closely resemble real images, making the image pairs realistic enough to hinder the discriminator's judgment. We can formulate this adversarial process as the following optimization problem:

$$\min_{R,G} \max_{D_J} \Big\{ \mathcal{L}_{\text{gan}}(R, G, D_J) = \mathbb{E}_{(\mathbf{X},\mathbf{Y})}\left[D_J(\mathbf{X}, \mathbf{Y})\right] - \alpha \mathbb{E}_{(\hat{\mathbf{X}},\mathbf{Y})}\left[D_J(\hat{\mathbf{X}}, \mathbf{Y})\right] - (1-\alpha)\mathbb{E}_{(\mathbf{X},\hat{\mathbf{Y}})}\left[D_J(\mathbf{X}, \hat{\mathbf{Y}})\right] \Big\} \quad (4)$$

where $\hat{\mathbf{X}} = R(\mathbf{Y})$ and $\hat{\mathbf{Y}} = G(\mathbf{X}, \mathbf{Z})$. The hyperparameter α determines the relative importance between G and R. Our optimization objective follows the Wasserstein-GAN [71] strategy, essentially using the Wasserstein-1 distance as the distance measure between distributions.

Through this joint adversarial process, the generator and the denoiser mutually promote each other by their dual regularization. Compared to conditional modeling, modeling the joint distribution more effectively utilizes the complete information underlying the data. Specifically, given any pseudo pair $(\mathbf{X}, \hat{\mathbf{Y}})$ from generator G or $(\hat{\mathbf{X}}, \mathbf{Y})$ from denoiser R, the joint discriminator D_J is further trained against the real pair. Then, D_J is fixed as a criterion to update both R and G simultaneously. In this way, R and G can be jointly trained with the assistance of D_J, thereby establishing an implicit connection between noise generation and noise reduction. This collaborative mechanism encourages R and G to produce more realistic, clean, and noisy images simultaneously. From the perspective of generation, the denoiser plays a crucial role by providing additional fake samples that contribute to the adversarial interaction between the generator and the discriminator. These fake samples extra promote the update of the discriminator and thus indirectly facilitate the optimization of the generator.

2) Noise Distribution Approximation: The joint discriminator primarily discriminates the combined information of noise and image contents. However, due to the numerous sources of noise in HSIs, the noise exhibits local isotropy and global anisotropy across both spatial and spectral dimensions, resulting in the overall distribution of complicated noise. Therefore, we introduce an extra HSI noise discriminator D_N to implicitly discriminate between the real and the generated noise distributions, i.e., $p(\mathbf{N})$ and $p_G(\mathbf{N})$, and then push $p_G(\mathbf{N})$ toward the real noise distribution $p(\mathbf{N})$ in an adversarial fashion. Such a distribution alignment operation in the noise domain can help the training of the noise generator converge better under the complex HSI noise scenarios. The corresponding optimization problem can be formulated as follows:

$$\min_{G} \max_{D_N} \Big\{ \mathcal{L}_{\text{gan}}(G, D_N) = \mathbb{E}_{\mathbf{N}\sim p(\mathbf{N})}\left[D_N(\mathbf{N})\right] - \mathbb{E}_{\hat{\mathbf{N}}\sim p_G(\mathbf{N})}\left[D_N(\hat{\mathbf{N}})\right] \Big\} \quad (5)$$

where $\mathbf{N}$ denotes the real noise map obtained by $\mathbf{N} = \mathbf{Y} - \mathbf{X}$, and $\hat{\mathbf{N}}$ denotes the generated noise map by $\hat{\mathbf{N}} = \hat{\mathbf{Y}} - \mathbf{X}$.

3) Additional Constraints for Stabilizing the Adversarial Training: Previous research studies have highlighted the significance of traditional losses in accelerating and stabilizing the training of GANs [72], [73]. These losses play a crucial role in constraining the generated output, ensuring its alignment with the characteristics of real images. For noise removal, we adopt the L_1 loss as the reconstruction loss

$$\min_{R} \|\hat{\mathbf{X}} - \mathbf{X}\|_1. \quad (6)$$

For noise generation, directly measuring the distance between generated noise and real noise using the reconstruction loss is inappropriate, since we are more interested in making the two types of noises close with respect to their pixel-wise statistics than the original values. Therefore, to this aim, we introduce a pixel-wise statistical constraint of noise

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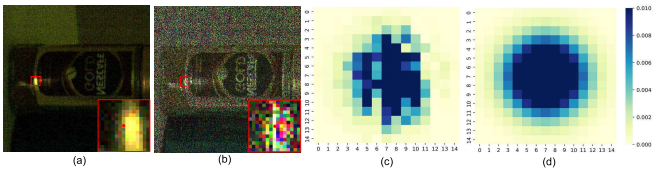


Fig. 2. Difference between Gaussian and bilateral filters. (a) Noisy HSI with bands 7, 15, and 23 from real dataset [15]. (b) Noise map corresponding to (a). (c) Heatmap of the bilateral filter kernel conducted within the red square in (b), where $k = 15$. (d) Heatmap of the Gaussian filter kernel. The noise shows an explicit SD characteristic, and the high score region in the bilateral filter kernel is consistent to the region associated with the red dot noise signal.

based on the KL divergence. The core idea is assuming that the noise in each pixel is drawn from a Gaussian distribution [74], and minimizing the KL divergence between the generated one and the real one

$$\min_{G} KL\left(\mathcal{N}\left([\hat{m}], \text{diag}([\hat{\sigma}^2])\right) \| \mathcal{N}\left([m], \text{diag}([\sigma^2])\right)\right) \quad (7)$$

where $[\cdot]$ denotes the reshape operation from matrix to vector, and $\hat{m}/m$ and $\hat{\sigma}^2/\sigma^2$ are the estimated mean and variance value map of the generated/real noise, respectively. Note that, though it seems simple, the pixel-wise Gaussian assumption is indeed very flexible, since the mean and variance vary with respect to pixels. It should also be mentioned that in [16], Yue et al. also constrained the noise, but they only considered the first-order statistical information. As a comparison, in this work, we adopt the KL divergence to simultaneously consider both the first- and second-order statistics of the HSI noise, due to its more complex structures.

The next question is how to estimate the mean and variance for the noise in each HSI pixel, since only the noise maps are available for both the generated and real HSIs. A common strategy is to use filters to average the pixel values within the neighborhood of the concerned pixel to get the estimations of its statistics. For example, Gaussian filtering was used in [16]. However, Gaussian filtering is not aware of brightness similarity, thus it ignores the signal dependency of HSI noise. Alternatively, bilateral filtering is an effective technique that not only smoothes the pixels within a window similar to Gaussian filtering, but also preserves edge information, as clearly shown in Fig. 2. Therefore, we adopt bilateral filtering to estimate the mean and variance for the noise of each pixel. Specifically, the means of the generated and the real HSI noise are estimated by

$$\hat{m} = \mathcal{BF}(\hat{\mathbf{Y}} - \mathbf{X}), \quad m = \mathcal{BF}(\mathbf{Y} - \mathbf{X}) \quad (8)$$

where $\mathcal{BF}$ denotes the bilateral filtering process, and then, the variances are estimated by

$$\hat{\sigma}^2 = \mathcal{BF}[(\hat{\mathbf{Y}} - \mathbf{X}) - \hat{m}]^2, \quad \sigma^2 = \mathcal{BF}[(\mathbf{Y} - \mathbf{X}) - m]^2. \quad (9)$$

4) Overall Training Objective: Integrating all the objectives discussed above, we obtain the final objective

$$\min_{R,G} \max_{D_I} \mathcal{L}_{\text{gan}}(R, G, D_I) + \min_{G} \max_{D_N} \mathcal{L}_{\text{gan}}(G, D_N) + \tau_1 \|\hat{\mathbf{X}} - \mathbf{X}\|_1 + \tau_2 KL\left(\mathcal{N}\left([\hat{m}], \text{diag}([\hat{\sigma}^2])\right) \| \mathcal{N}\left([m], \text{diag}([\sigma^2])\right)\right) \quad (10)$$

where τ_1 and τ_2 are hyperparameters to represent the weights of different losses.

C. Network Architectures

In this part, we discuss the network architectures within our proposed ADNG-Net. We will mainly focus on the design of the generator that approximates the conditional distribution $p_G(\mathbf{Y} \mid \mathbf{X}, \mathbf{Z})$, as well as its input. Then, we will briefly discuss other components that mostly follow existing settings.

1) Generator Architecture: The noise generator primarily extracts noise-related features from the clean HSIs and integrates them with the latent random variable $\mathbf{Z}$ to generate the noisy HSI $\mathbf{Y}$. The main design mechanism of the noise generator is trying to capture the local–global correlation and spatial signal dependency of the HSI noise. Therefore, we consider combining the transformer structures with convolutions, which have shown their effectiveness in capturing the global and local correlations, respectively. Specifically, as shown in Fig. 3, our generator network is constructed based on a UNet structure, which takes the concatenation of the clean HSI and the latent variable $\mathbf{Z}$ as input, and outputs the noisy HSI by adding the generated SD noise to the clean HSI. Within the UNet structure, each of the encoder's first three layers includes a local-feature-aware (LFA) block to integrate HSI local features with noise, and an LN layer for stable training. The encoder's final layer uses a global-feature-aware (GFA) block to fuse noise with global image features.

The LFA block is essential for deep feature fusion, extracting local spatial correlations from the clean HSI and fusing them with noise. This block combines the transformer and convolutional operations. Specifically, we use a Swin transformer layer (SWTL) [36] with window-based multi-head self-attention (W-MSA) and shifted W-MSA (SW-MSA) applied successively on the spatial dimension, as shown in Fig. 3(a). The shifted windows enhance connections between local windows, allowing for better interaction of locally fused features. As a result, SWTL improves the network's ability to encode local interactions and distant dependencies efficiently. In addition, a convolution layer at the end of each block enhances the translation invariance of features.

After applying three local-feature-aware blocks (LFA-blocks) to the input, the raw features of the original HSI and the noise map are fused at multiple scales. Each point in the feature map has a certain receptive field of the original image. To further capture the global correlation of the encoded high-level semantic features, we employ the GFA block, which comprises a global-window-based transformer layer (GWTL) and a convolution layer. As shown in Fig. 3(b), GWTL is composed of two transformer sublayers that incorporate global-window-based multihead self-attention (G-MSA), with the window as the same size of the feature map. By attending to the entire map, the G-MSA enables the model to further extract the feature of HSIs globally and meanwhile fuse it with noise.

2) Input to Generator: As discussed in Section III-A.1, the random input $\mathbf{Z}$ should better be dependent on the clean HSI $\mathbf{X}$ to help generate the realistic HSI noise. In this part, we discuss how to make such a dependency explicit. Besides, we also briefly discuss the preprocessing of $\mathbf{X}$.

According to previous research studies [27], [30], [75], the noise levels in different bands of HSI exhibit different patterns

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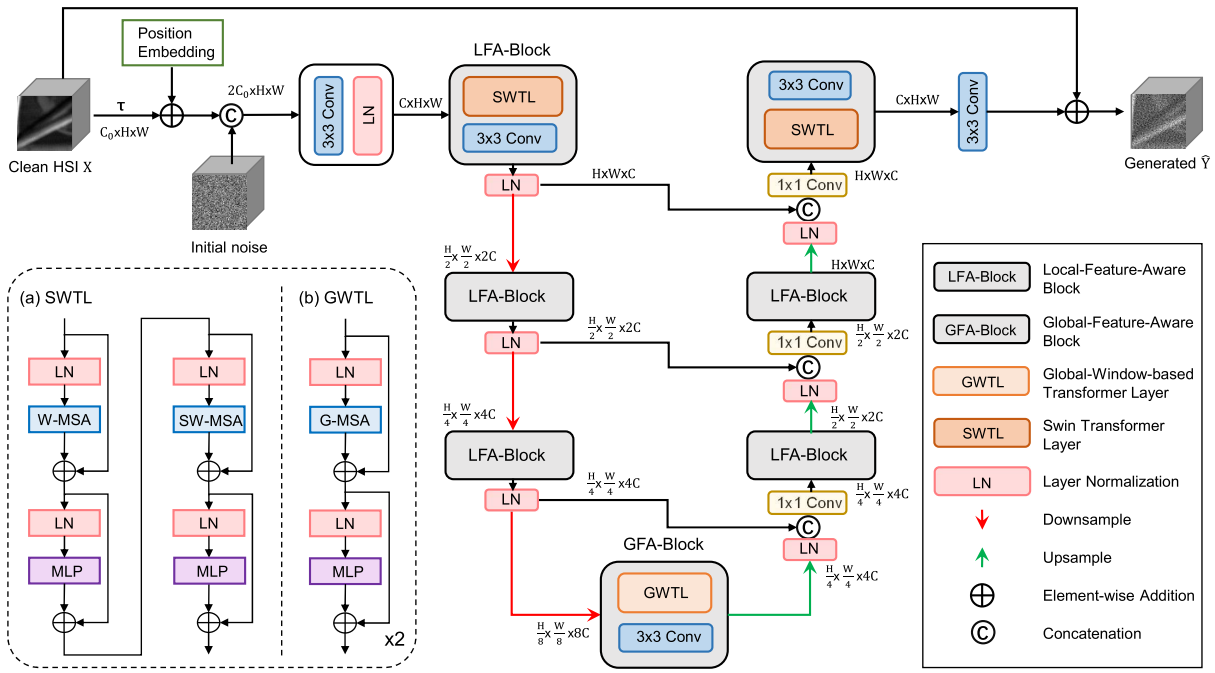


Fig. 3. Architecture of the noise generator. The U-shaped network is the main part of the generator, which consists of LFA-Block for the local feature fusion and GFA-Block to combine the global correlation. Core layers are based on transformers. (a) SWTL. (b) GWTL.

or rules. Therefore, we embed the knowledge of spectral difference into the latent variable $\mathbf{Z}$. Specifically, we assume that each band of $\mathbf{Z}$ is nonidentical and independently sampled from a Gaussian distribution with zero mean and variance reflecting the noise intensity of the band. Since we assume the HSI noise is signal dependent, these variances can be predicted from the original clean HSI. Therefore, $\mathbf{Z}$ can be drawn by the following process:

$$\begin{aligned} \mathbf{Z}_i &\sim \mathcal{N}(0, \sigma_i^2) \\ \sigma_i^2 &= f(\text{scaled}(\overline{\mathbf{X}_i})) \end{aligned} \tag{11}$$

where $Z_i (i = 1, 2, \ldots, B)$ denotes the ith band of the input $\mathbf{Z}$, σ_i^2 is the variance of the input, $\overline{\mathbf{X}_i}$ denotes the average intensity of the original image in band i, scaled$(\cdot)$ denotes the normalization process of the spectral average intensity to scale $\overline{\mathbf{X}_i}$ between 0 and 1, and the function f is a multilayer perceptron (MLP). Then, the next question is how to specify the supervision label to train this MLP, which is a little bit tricky since no ground-truth noise intensity is available. Nevertheless, we can empirically compute the sample variance of the noise of each band from the training data, and let it to be the output goal of the MLP. Though looks heuristic, such a strategy works well in our experiments.

For the other input, i.e., the clean HSI $\mathbf{X}$, to maintain a similar numerical level to the latent variable $\mathbf{Z}$, we multiply it by a scaling ratio τ. Furthermore, we incorporate position embedding to enhance the representation of spatial information. The scaled and position-coded clean image is then concatenated along the spectral segments with the latent variable as the input of the generator.

3) Other Components: We choose classical architectures for the denoiser and two discriminators. For the denoiser, we employ the HSID-CNN structure [27] with multiscale feature extraction and multilevel feature representation. It should be noted that our ADNG-Net is designed to be general and does not rely on a specific denoiser. Therefore, other network architectures can also be employed as the denoiser in ADNG-Net.

For the discriminators, we employ architectures similar to the discriminator in PatchGAN [72], while modifying the size of the output score map. Each score represents the judgment of a small area of the input image.

IV. Experiments

In this section, we conduct a series of experiments to comprehensively evaluate the effectiveness of the proposed method. We first discuss the implementation details for training the proposed ADNG-Net on the paired HSI dataset collected by Zhang et al. [15]. Then, we provided experimental results on both this paired HSI dataset and real-world HSI datasets. At last, we conduct ablation studies to show the necessity of each component in our framework.

A. Implementation Details for Training ADNG-Net

1) Hyperparameters of Subnetwork Architectures: For our designed generator, after normalizing the clean HSI data, the magnitude of the clean input $\mathbf{X}$ is much higher than that of $\mathbf{Z}$; therefore, we empirically set τ to 0.01, which was found to work well for the training. The number of channels is adjusted to $C = 128$ after the shallow feature extraction, the window size of SWTL is 8, and the numbers of heads are set as $2, 4, 4$, and 8 from the topmost block to the bottommost block, respectively. The joint discriminator uses three convolution

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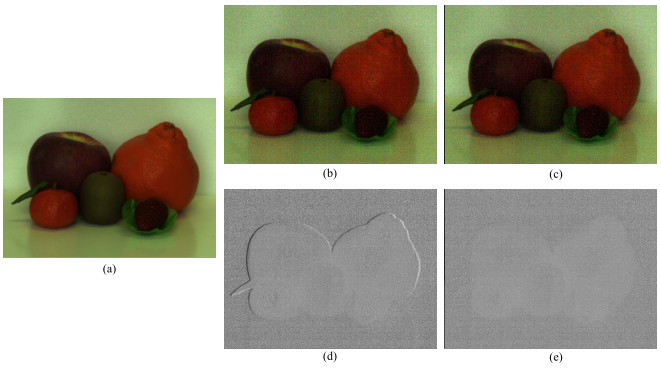


Fig. 4. Misalignment example of the realistic dataset. (a) Clean HSI. (b) Original noisy HSI. (c) Aligned noisy HSI. (d) Original noise map. (e) Aligned noise map.

layers with 3×3 convolutional kernels and a stride of 1, while the noise discriminator uses two convolution layers with 4×4 convolutional kernels and a stride of 2. For the HSID-CNN, the number of adjacent bands is set as 24, and we use the default parameters for the other components of it.

2) Dataset and Preprocessing: We use the Realistic Dataset [15] for training our ADNG-Net. The dataset consists of 59 noisy HSIs and their corresponding clean images, which were captured by the SOC710-VP hyperspectral camera. It includes indoor and outdoor scenes, with each HSI having a spatial resolution of 696×520 and 34 spectral bands ranging from 400 to 700 nm.

The original dataset contains various issues, such as global and local misalignment, image rotation, and color bias. We show a misalignment example in Fig. 4. As can be seen, according to the noise map obtained by calculating the residual between the noisy image and its corresponding clean one, there is a slight shift between these two images. Therefore, we perform a preprocessing to improve the data quality, which involves performing image alignment, filtering, and cropping. As an example, Fig. 4(c) and (e) shows that by inversely shifting the noisy HSI, it is well-aligned with its clean counterpart. After preprocessing, we obtain 30 high-quality "clean–noisy" HSI pairs with different sizes. Since the spectral value ranges from 0 to 4095, we normalize each image by dividing the pixel value by 4096, while preserving the difference in spectral signal intensity. Four HSIs (one indoor and three outdoors) are selected for testing, while the rest are used for training. To increase the diversity of training samples, we crop each image into spatial overlapping patches with a sliding window, whose size is $34 \times 64 \times 64$. To further purify the dataset, we remove image pairs with large color bias from the cropped dataset. Random flipping and rotation are further applied to augment the purified dataset.

3) Training Algorithm and Hyperparameter Settings: During the training stage of ADNG-Net, we have four networks to train: the denoiser R, the generator G, the joint discriminator D_J, and the noise discriminator D_N. The four networks are optimized in an alternating fashion. Specifically, for every training cycle, three batches are used to update the discriminators D_J and D_N, while keeping the generator G and the denoiser R fixed. Then, with the discriminators fixed as the criterion, one batch is used to update R and G separately. These two steps are repeated iteratively throughout the training process. The overall training procedure is demonstrated in Algorithm 1. We configure the Adam optimizer to train the four, with momentum terms of (0.9, 0.999) for R and (0.5, 0.9) for G, D_J, and D_N. The initial learning rate is set as 5e-5 and 1e-5 for R and G, respectively, and 2e-4 for D_J and D_N. The learning rates decay exponentially with a factor of 0.95. The networks undergo training for 70 epochs using a batch size of 2.

Algorithm 1 Training Procedure of ADNG-Net

Require: batch size, τ_1, τ_2, α
Initialize networks: Generator G, Denoiser R, Joint Discriminator D_J, Noise Discriminator D_N
Set iteration counter $i \leftarrow 1$
while not converged **do**
  Sample a batch of pairs $(\mathbf{X}, \mathbf{Y})$ from $p(\mathbf{X}, \mathbf{Y})$
  Sample a batch of pairs $(\hat{\mathbf{X}}, \mathbf{Y})$ from $p_R(\mathbf{X}, \mathbf{Y})$ and $(\mathbf{X}, \hat{\mathbf{Y}})$ from $p_G(\mathbf{X}, \mathbf{Y})$
  Update D_J and D_N using current batch, keeping G and R fixed
  if i mod 3 = 0 **then**
    Update R with fixed D_J, D_N and G
    Update G with fixed D_J, D_N and R
  end if
  Increment $i \leftarrow i + 1$
end while

For the objective in (10), τ_1 is set as 1000 and τ_2 is set as 100. For the adversarial loss in (4), α is set as 0.5, meaning the denoiser and the generator contribute equally to the generation of fake image pairs. For the bilateral filter in (8) and (9), the kernel size K is set as 15.

B. Results on Paired Dataset

On the paired dataset, we evaluate the effectiveness of the proposed method from both the noise generation and noise removal perspectives. For noise generation, we show whether the generator is able to generate realistic HSI noise, and for noise removal, we verify if the generated noisy HSIs can enhance the performance of denoisers.

1) Noise Generation: To quantitatively evaluate the generated noise, we adopt three metrics. The first is the KL divergence between the real noise and the synthetic fake noise, which is considered as samples drawn from the real noise distribution $p(\mathbf{N}^r)$ and the fake noise distribution $p(\mathbf{N}^f)$, respectively. The second is average KL divergence (AKLD) [16], which utilizes the pixel-wise Gaussian assumption for the noise, and then measure the KL divergence between $p_{\mathbf{Y}^f}(\mathbf{Y} \mid \mathbf{X})$ on the fake image pairs $(\mathbf{X}^r, \mathbf{Y}^f)$ and $p_{\mathbf{Y}^r}(\mathbf{Y} \mid \mathbf{X})$ on the real image pairs $(\mathbf{X}^r, \mathbf{Y}^r)$. For the calculation of KL and AKLD, we randomly generate 100 noisy images for a single clean image and take the average to reduce the influence of

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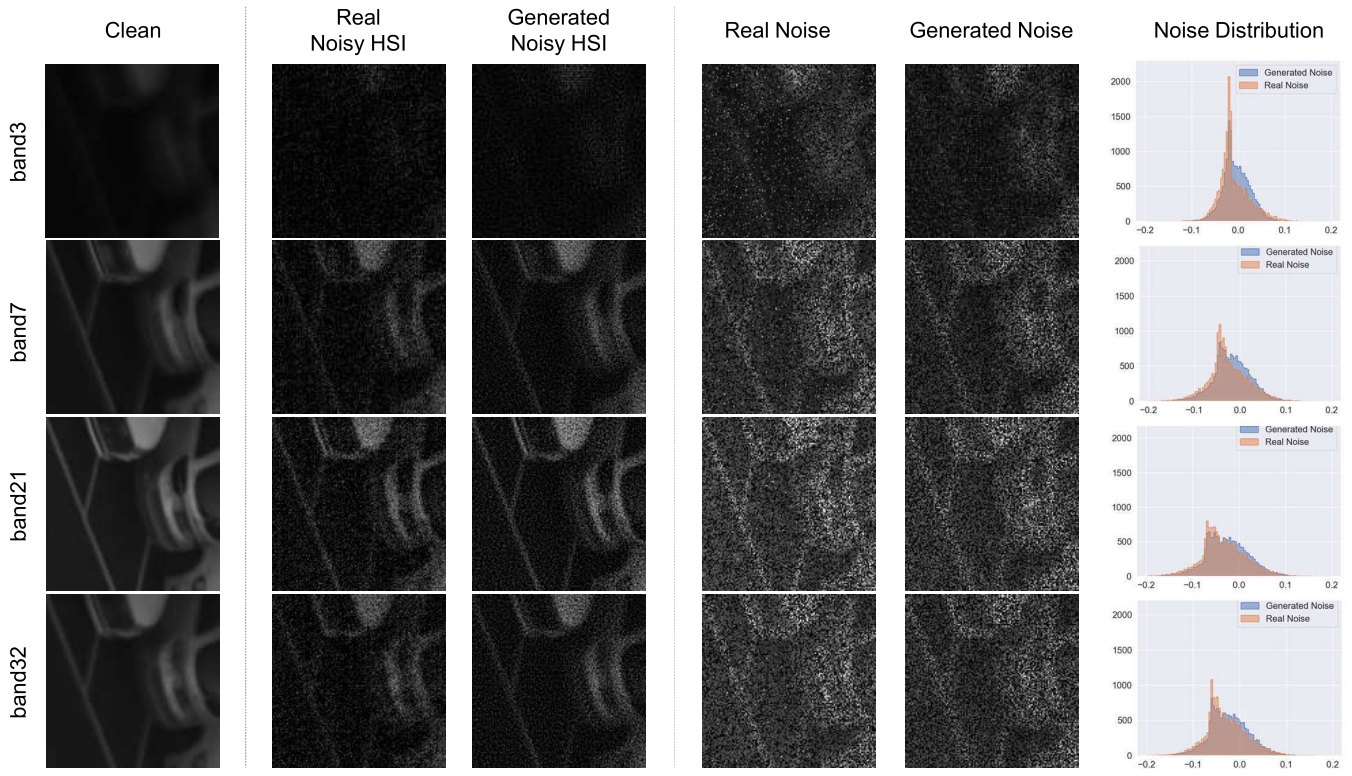


Fig. 5. Visualization of the generated noise of bands 3, 7, 21, and 32 in image 48 in test data of the Realistic Dataset [15]. Patches of 128×128 are chosen for comparison. Both the noisy image and the noise map are shown for the real HSI and the generated one. The noise distributions of the real HSI and the generated one in different bands are also shown.

TABLE I
QUANTITATIVE EVALUATION OF NOISE GENERATION RESULTS ON THE REAL DATASET

Metrics	Complex	Homo.	Hetero.	Ours
PGAP ↓	3.62	4.59	4.47	**0.36**
AKLD ↓	2.310	0.062	0.101	**0.059**
KL ↓	0.173	0.058	0.059	**0.035**

randomness, i.e.,

$$\mathrm{KL} = \frac{1}{L}\sum_{j=1}^{L} KL\left[p_{\mathbf{N}^{f_j}}(\mathbf{N}) \| p_{\mathbf{N}^r}(\mathbf{N})\right],$$

$$\mathrm{AKLD} = \frac{1}{L}\sum_{j=1}^{L} KL\left[p_{\mathbf{Y}^{f_j}}(\mathbf{Y} \mid \mathbf{X}) \| p_{\mathbf{Y}^r}(\mathbf{Y} \mid \mathbf{X})\right] \quad (12)$$

where $L = 100$. The third metric is PGAP [16], which measures the gap between the PNSR values obtained by the denoisers trained on the real data and generated data, respectively. We train two R_1 and R_2, respectively, on the real data and the generated fake data, then test the denoiser on the same test set Λ, leading to the following PGAP:

$$\mathrm{PGAP} = \mathrm{PSNR}(R_1(\Lambda)) - \mathrm{PSNR}(R_2(\Lambda)). \quad (13)$$

We first list some quantitative results on the generated noise in Table I, where we compare our method with other noise synthesis methods, including the commonly used complex noise model [4], the homoscedastic Gaussian noise model, and the heteroscedastic Gaussian noise model [61]. For the calculation of PGAP, the test set of the Real Dataset is regarded as Λ, and HSID-CNN [27] is selected as the denoiser. Apparently, our proposed method achieves the best performance. Taking the PGAP metric as an example, the difference between the results obtained by training on our generated data and on the real data is negligible. This indicates that even without using real data and solely relying on the generated data, we can still achieve a highly promising denoising performance.

We then show some qualitative results of noise generation. We select several bands to show their real noisy images, noise maps, and noise distribution, respectively, as illustrated in Fig. 5. For the noise map, we take the absolute value to show the intensity of the noise and multiply it by 5 to enhance visibility. It can be seen that the noise generated by our method effectively synthesizes the spectrum correlation and difference. The generated noise and the real noise exhibit similar properties and distribution characteristics in each spectrum. It is also worth noting that under the guidance of the original image, the noise we generate successfully preserves the signal dependency.

To further validate the fidelity of our generated noise, we compare the noise generation results of our method with other noise synthesis methods in Fig. 6. The complex noise model shows clear differences from real noise, and the i.i.d. Gaussian noise model fails to capture spectral variations of noise, leading to visible color bias in the noise map. Both heteroscedastic

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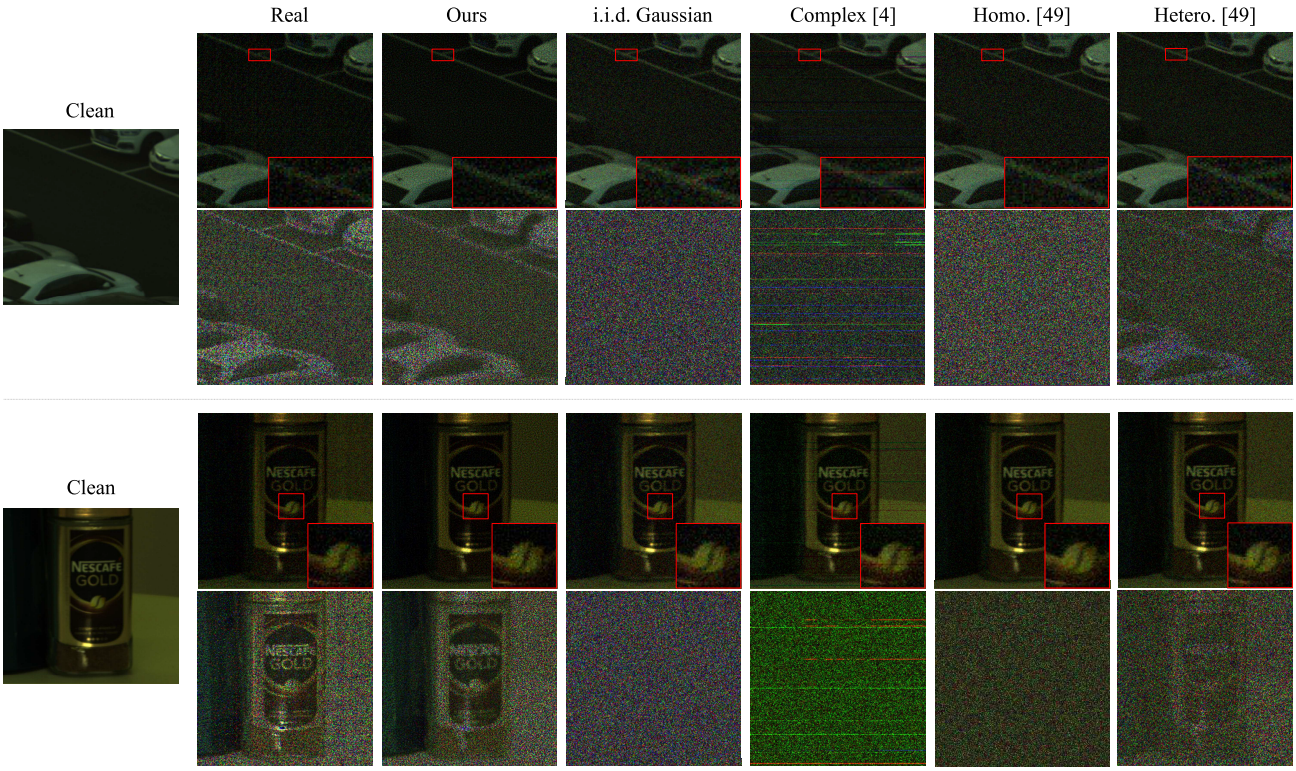


Fig. 6. Noise generation results by different methods on images 38 and 48. Patches of 256 × 256 are chosen for comparison. For each clean image patch, the first row shows the noisy images, and the second row shows the corresponding noise maps.

TABLE II

DENOISING RESULTS COMPARISONS ON THE REAL DATASET. "+" DENOTES THE AUGMENTED TRAINING. "△" DENOTES THE PERFORMANCE GAIN BY THE AUGMENTED TRAINING

Metrics	Noisy	DeNet	DeNet+	△	HSID-CNN	HSID-CNN+	△	QRNN3D	QRNN3D+	△	T3SC	T3SC+	△	SERT	SERT+	△
PSNR ↑	24.79	30.67	35.57	+4.90	36.61	37.07	+0.46	36.65	37.22	+0.57	37.34	37.61	+0.27	38.13	39.03	+0.90
SSIM ↑	0.356	0.781	0.942	+0.161	0.959	0.963	+0.004	0.945	0.953	+0.008	0.961	0.963	+0.002	0.958	0.966	+0.008
SAM ↓	0.474	0.213	0.089	-0.124	0.065	0.062	-0.003	0.079	0.076	-0.003	0.072	0.066	-0.006	0.070	0.059	-0.011
ERGAS ↓	292.0	149.6	77.9	-71.70	69.60	65.86	-3.74	70.28	65.77	-4.51	64.92	63.12	-1.80	61.60	54.88	-6.72

and homoscedastic Gaussian models struggle to capture the signal dependency of noise. In contrast, our method generates noise that is most visually close to the real one. Specifically, in areas where the image signal exhibits more pronounced variations, our generated noise closely resembles the real noise, and other methods tend to make the noisy regions with high signal intensity overly clear, while introducing excessive noise in low-intensity areas such as the background.

2) Noise Removal: As discussed before, the ultimate goal of noise generation is to enhance the performance of HSI denoisers under the scenario with limited data, by including the generated "clean–noisy" HSI pairs in the training process. To verify this effect of the proposed generator, we conduct experiments with two ways of using the generated data: data augmentation and fine-tuning. The former directly augments the training set with the generated data, following the way in [16]. Specifically, the original training set contains 5728 image pairs, and we augment it by generating 4× that amount of additional data. The latter fine-tunes the pretrained denoiser on extra data that includes a portion of the generated ones [76]. Four standard metrics are adopted to evaluate the denoising performance: the peak signal-to-noise ratio (PSNR), the structural similarity (SSIM) [77], the spectral angle mapper (SAM) [78], and the normalized global error in synthesis (ERGAS) [79].

We first show the results of the data augmentation setting. Specifically, we compare the performance of a denoiser trained only on the real data, with that of trained on the augmented data combining both the real and generated ones. We consider five representative denoisers, including DeNet [26], HSID-CNN [27], QRNN3D [34], T3SC [32], and SERT [80]. Among them, DeNet and HSID-CNN utilize 2-D convolutional networks, while QRNN3D combines a quasi-recurrent pooling structure with 3-D convolution. T3SC employs a learnable sparse coding model, and the more recent SERT is a transformer-based denoiser for HSI. The results with respect to the aforementioned four metrics are summarized in Table II. It can be seen that, with the augmented training data, all

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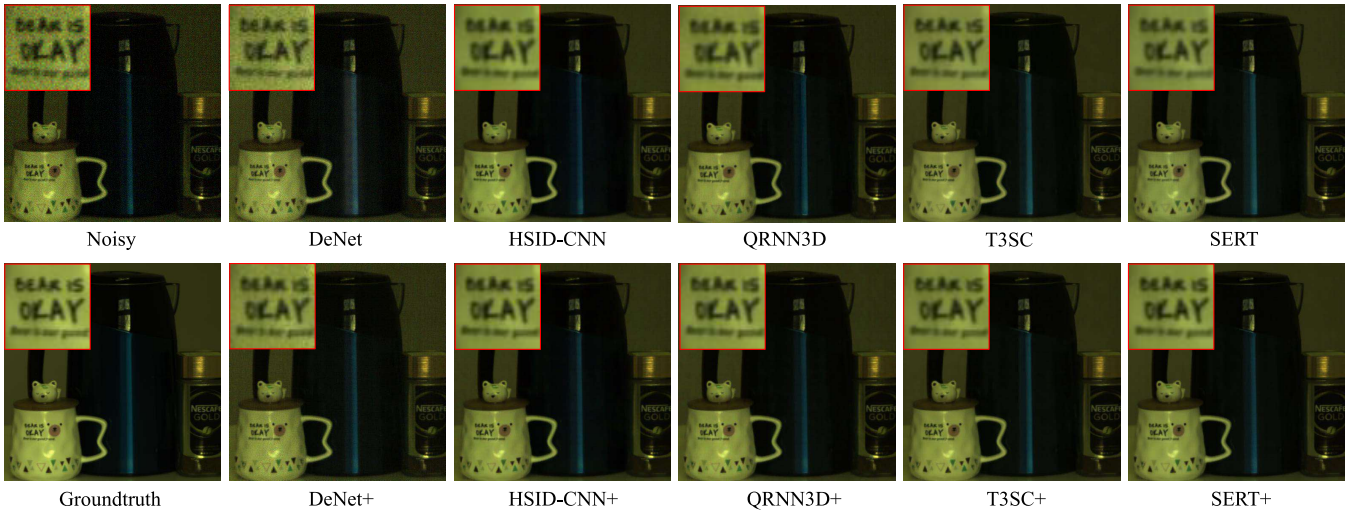


Fig. 7. Visual comparisons on a 512 × 512 patch of the test image 48, including denoising results of different denoisers trained on the original (first row)/augmented (second row) Realistic Dataset.

denoisers exhibit significant improvements in their denoising performances. In particular, for the simplest network, DeNet, the augmentation of the training set substantially enhances its performance, near 5 dB in PSNR. For other denoisers with more complex structures, the augmented training set can still improve the performance. For instance, SERT, which already has the best performance with the original training data, obtains a 0.9-dB improvement in PSNR and a 6.72-dB improvement in ERGAS. It is important to note that we did not specifically tailor the training strategy for any denoiser. Therefore, we can conclude that the generated data serves as a robust augmentation of the original training set, regardless of the denoiser. This finding strongly supports that the noise generated by our generator possesses similar characteristics to the real noise. Fig. 7 further shows the visual denoising results of different denoisers before and after data augmentation. It can be clearly observed that after training on the augmented dataset, all the denoisers produce denoised images that are noticeably clearer and exhibit reduced chromatic aberration.

Then, we show the results of using the generated data and the Realistic Dataset training data jointly to finetune HSID-CNN, which is pretrained on the real data. Specifically, we vary the ratio of the generated data to the real data, denoted as q, to observe the impact of the generated data on finetuning. The results are summarized in Table III. It is observed that the denoising performance slightly decreases when we further train the denoiser using only the real data from the same training set ($q = 0$), indicating the potential overfitting of the denoiser on the training data. However, as we gradually increase the proportion of the generated data mixed in during fine-tuning, the denoising performance steadily improves. The best performance is achieved when $q = 0.8$, resulting in an increase of 0.14 in PSNR. This suggests that the generated data distribution closely resembles the distribution of real data, enabling the denoiser to further learn the real noise characteristics from these data. While the generated data differs from real data to some extent, a slight decline in performance is observed when q surpasses 0.8. Nonetheless, the performance remains higher than the pretraining results, which demonstrates that even with a large amount of generated data, fine-tuning can still enhance the denoising performance.

TABLE III
FINETUNING RESULTS WITH DIFFERENT PROPORTIONS OF THE GENERATED DATA

q	PSNR ↑	SSIM ↑	SAM ↓	ERGAS ↓
Pretrained	36.61	0.9588	0.0650	69.60
0	36.52	0.9581	0.0652	70.21
20%	36.67	0.9589	0.0644	68.31
40%	36.69	0.9593	0.0644	68.52
60%	36.69	0.9593	**0.0642**	68.14
80%	**36.75**	**0.9593**	0.0645	**67.83**
100%	36.73	0.9590	0.0647	68.04
120%	36.71	0.9584	0.0655	68.27

C. Results on Real-World Dataset

Now, we show the effectiveness of the proposed method in helping deal with real-world HSI denoising. Specifically, since the noise level and characteristics of real-world HSIs are generally unknown, it is a common practice to synthesize noise (e.g., i.i.d./non-i.i.d. Gaussian noise) on clean HSIs to generate paired data for training the denoiser. Similarly, in our evaluation, we employ our trained generator as a noise-synthesizing tool to generate "clean–noisy" training pairs. For a comprehensive evaluation, we consider different ways to construct the training data, including only the real paired data, the augmented data with both real paired data and the generated ones, and the pure synthetic data obtained by different noise-adding approaches (details are provided later). For the synthetic training set, we select about 7200 clean patches with the spatial size 64×64 from the Realistic Dataset as well as 10 000 from Washington DC Mall image, which

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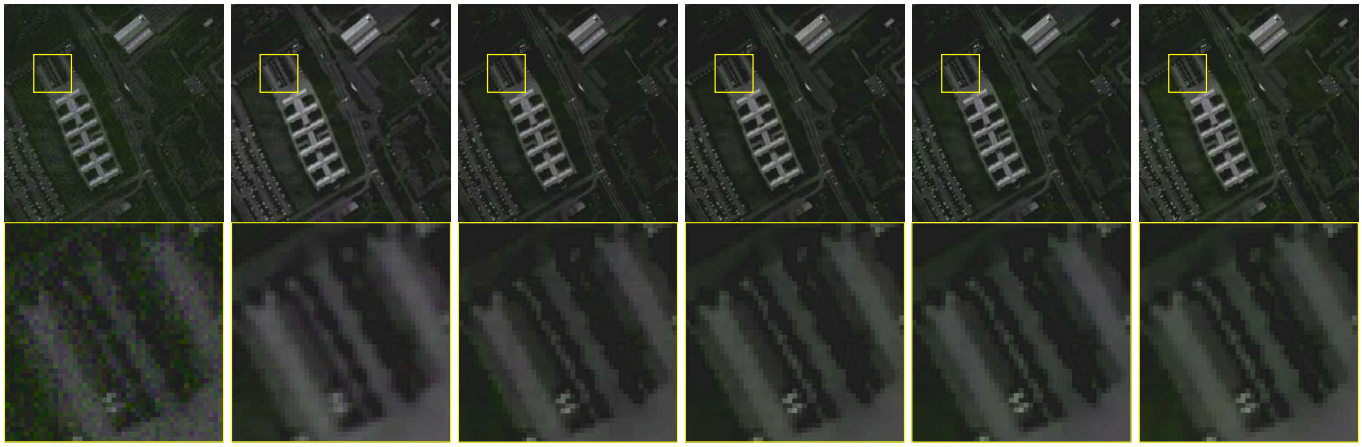


Fig. 8. Visual comparison of denoising results with bands 2-1-3 as R-G-B on Pavia University Dataset. (a) Original noisy image. (b)–(f) Results of denoisers trained on training sets constructed in different ways. (b) Real Pairs. (c) i.i.d. Gaussian. (d) Non-i.i.d. Gaussian. (e) Real+Generated. (f) Generated.

is obtained by the Hyperspectral Digital Imagery Collection Experiment airborne sensor.[1] We adopt HSID-CNN as the denoiser throughout this section, and consider the following two real-world HSI datasets.

1) *Pavia University [81]:* It is part of an HSI of the city of Pavia, Italy, taken in 2003 by the imaging system ROSIS-03. This image comprises 610×340 pixels and 115 bands. We crop it into 256×256.
2) *Indian Pines [82]:* It was collected by a 224-band AVIRIS sensor over Indian Pines test site in Northwestern Indiana. This image comprises 145×145 pixels and 206 bands (after removing water absorption bands).

1) Results on Pavia University: The first few bands in the Pavia University dataset are mainly corrupted by different degrees of Gaussian-type noise. Hence, we consider the following ways to construct the training set.

1) *i.i.d. Gaussian:* All the training data are synthesized by corrupting the clean HSIs with additive Gaussian noise with zero mean and a specific variance.
2) *Non-i.i.d. Gaussian:* The training data are synthesized by corrupting the clean HSIs with Gaussian noise, whose mean is zero and variance varies randomly along the spectral band.
3) *Real pairs:* The training set is formed using the same paired data as the Realistic Dataset.
4) *Generated:* The noisy HSIs are generated by our trained ADNG-Net to form the "clean–noisy" training pairs.
5) *Real+Generated:* The training set is constructed combining the generated and the real pairs together, with the number of generated pairs four times of the real ones.

To ensure a fair comparison, we train the denoiser using the same training strategy. The denoising results are shown in Fig. 8. It can be seen that when using only *Real pairs* as the training set, the visual denoising effect is not very good, probably due to the limited amount of real data available. When the training data is synthesized by corrupting clean HSIs with Gaussian-type noise, the denoised images still exhibit some noise, leading to a lack of local smoothness in detail. Moreover, the colors are not correctly preserved; this is probably because the denoiser removes some image content due to a lack of signal dependency of the synthesized noise. When using *Real+Generated* as the training data, the denoising results demonstrate improved visual quality by preserving more texture and local smoothness while effectively removing noise. Comparably, training with only the generated noise achieves even better visual results, with its color closer to the original HSIs and more details preserved. We also present the spectral curve of denoised images in Fig. 9. As shown, the denoised HSI obtained by training with our generated noise best preserves the original spectral trend while achieving local smoothness. In summary, the above results suggest that our generator can produce a large number of realistic and effective "clean–noisy" pairs at a low cost, efficiently enhancing the denoising performance.

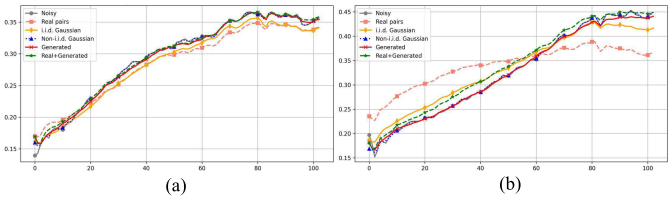


Fig. 9. Denoised spectral signatures by different training data of point (a) $(101, 155)$ and (b) $(127, 123)$ of Pavia University Dataset.

2) Results on Indian Pines: In Indian Pines dataset, the first few bands and several other bands are affected by various types of noise, including Gaussian, impulse, and stripe noises, which are more complex than those of Pavia University. Therefore, in addition to the aforementioned ways to construct the training set, we also consider the following training set with complex noise components in our analysis:

1) *Non-i.i.d. Gaussian+Stripe+Impluse:* Based on the non-i.i.d. Gaussian noise in the training set (2), we further simulate impulse noise at random intensities on one-

[1] https://engineering.purdue.edu/biehl/MultiSpec/hyperspectral.html

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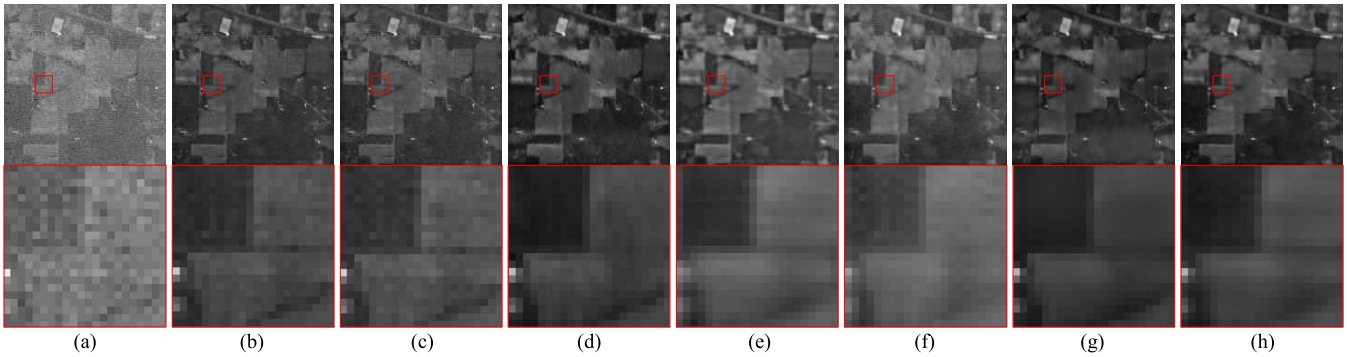


Fig. 10. Visual comparison of denoising results on Indian Pines Data. The original noisy image of band 2 is shown in (a). The denoiser is trained on different training sets, i.e., (b) i.i.d. Gaussian, (c) Non-i.i.d. Gaussian, (d) Non-i.i.d. Gaussian+Stripe+Impluse, (e) Real pairs, (f) Real+Generated, (g) Generated+Stripe+Impluse, and (h) Generated.

third of the bands. In addition, stripe noise, affecting 10%–50% of the rows, is randomly introduced to another third of the bands [4].

2) *Generated+Stripe+Impluse:* Since our generator does not focus on pattern noise as it can be simulated in a much simpler way [4], [34], [80], we introduce stripe and impulse noises similar to that in training set (6), on the basis of the generated noise in training set (4).

The denoising results of band 2 in Indian Pines Dataset are visualized in Fig. 10. It can be observed that training only with generated noise can already bring effective denoising results, successfully removing more noise than training with the data fully synthesized by traditional methods [4], [34], [35], as well as the limited real data, while maintaining image details. Moreover, combining our generated random noise with simulated stripe and impulse noise leads to further improvements in training performance. The advantage of our generated noise is also evident when comparing Fig. 10(d) and (g): replacing simulated non-i.i.d. Gaussian noise with our generated random noise results in a closer match to the real noise characteristics observed in Indian Pines dataset. This demonstrates the flexibility and extensibility of our framework: it generates realistic, complex random noise distributions while allowing easy integration with deterministic pattern noise, thereby enabling comprehensive modeling of real HSI noise.

For a more comprehensive comparison and to further show the effects of HSI denoising on downstream tasks, we also perform a classification experiment on Indian Pines Dataset, comparing results before and after denoising. The classification is conducted using a support vector machine (SVM) with ground truth labels of 16 categories. Totally, 20% samples from each category are randomly selected for training, and the rest for testing. The performance is quantified using three metrics: overall accuracy (OA), average accuracy (AA), and kappa coefficient (kappa) [83]. The obtained results are shown in Table IV. It is clear that the denoising process significantly improves classification accuracy. Among all the noise-adding methods, our approaches yield better results than those with synthetic noise. Besides, using noise generated by our method alone is already competitive enough regarding the classification results.

TABLE IV

CLASSIFICATION ACCURACY RESULTS FOR INDIAN PINES IMAGE USING SVM. THE BEST AND SECOND RESULTS ARE HIGHLIGHTED IN BOLD AND UNDERLINED, RESPECTIVELY. "S" DENOTES STRIPE NOISE AND "I" DENOTES IMPULSE NOISE

Metrics	AA	kappa	OA
Original	0.7676	0.8362	0.8571
i.i.d. Gaussian	0.9166	0.9711	0.9748
Non-i.i.d. Gaussian	0.9214	0.9560	0.9615
Non-i.i.d.-Gaussian+S+I	0.9094	0.9637	0.9683
Real Pairs	0.9374	0.9722	0.9757
Generated+S+I	0.9265	<u>0.9754</u>	<u>0.9785</u>
Real+Generated	**0.9390**	<u>0.9754</u>	<u>0.9785</u>
Generated	<u>0.9381</u>	**0.9768**	**0.9797**

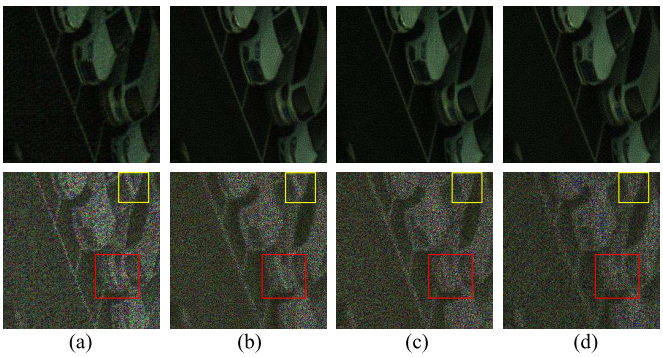


Fig. 11. Real or generated noisy images and the corresponding noise. (a) Real case. (b) Generated results with Reg_B. (c) Generated results with Reg_G. (d) Generated results with ADNG-Net_{SI}.

The results of Fig. 10 and Table IV jointly confirm that the noise generated by our method is realistic and diverse enough to improve the denoising performance on real noisy HSIs, as well as the performance of the downstream tasks.

D. Ablation Study

To further validate the rationality and effectiveness of some detail settings in our method, we conduct ablation experiments.

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TABLE V
COMPARISON RESULTS OF DIFFERENT SETTINGS ON REALISTIC DATASET TEST DATA

	D_J	D_N	Reg_G	Reg_B	PGAP ↓	AKLD ↓	KL ↓
Baseline	✓				2.51	0.174	1.339
Baseline + D_N	✓	✓			2.40	0.080	0.088
Baseline + Reg_B	✓			✓	2.18	0.044	0.566
Baseline + D_N + Reg_G	✓	✓	✓		1.00	0.051	0.050
ADNG-Net$_{SI}$	✓	✓		✓	0.83	0.063	0.044
ADNG-Net	✓	✓		✓	0.36	0.059	0.035

TABLE VI
COMPARISON RESULTS OF BASEG AND ADNG-NET ON REALISTIC DATASET TEST DATA

Metrics	Methods	
	BaseG	ADNG-Net
PGap ↓	0.44	0.35
KL ↓	0.046	0.035

TABLE VII
GENERATED RESULTS OF ADNG-NET TRAINED WITH SEMI-SUPERVISED STRATEGY USING 1/2 AND 1/4 PAIRED SAMPLES

r	PGAP↓	AKLD↓	KL↓
1	0.36	0.059	0.035
1/2	0.5	0.054	0.038
1/4	0.64	0.047	0.041

All experiments in this part are carried out on the Realistic Dataset.

1) Impact of Loss Settings: In addition to the joint adversarial loss inherited from [16], the training of the noise generator benefits from the newly designed noise adversarial loss defined in (5), which is provided by the noise discriminator D_N, as well as the generator's regularizer through bilateral filtering (Reg$_B$) defined in (7). We show the effectiveness of these elements in Table V, where we denote the model trained without D_N and Reg$_B$ as baseline, the generator's regularizer through Gaussian filtering as Reg$_G$. It can be observed that both the noise discriminator and the regularizer contribute to enhancing the quality of noise generation. It should be mentioned that although the AKLD value of using ADNG-Net is slightly higher than that of only using Reg$_B$ with baseline, this index only evaluates the local statistics of noise within a small window and cannot measure the characteristics of noise globally. Therefore, according to PGAP and KL metrics, we can conclude that generally, the quality of the noise generated under all the loss functions we adopted is found to be superior. Then, we further compare the filtering effects between Gaussian filtering and bilateral filtering. As can be seen from Table V, the results of using Reg$_B$ are better than those of using Reg$_G$, with respect to PGAP and KL, showing the superiority of the bilateral filtering. We also visualize the noise generation results for using Reg$_G$ and Reg$_B$ in Fig. 11, and the local details within the yellow and red boxes clearly show that the noise regularizer with bilateral filtering more effectively preserves the SD noise component in HSI noise.

2) Impact of the Denoiser: Intuitively, the noise generator can be trained without the denoiser. However, the training of the denoiser is indeed crucial in the dual-adversarial framework to promote the training of the generator. To verify this, we train ADNG-Net without the denoiser and denote the trained model as BaseG. The comparison results of these two models on noise generation are listed in Table VI. It is evident that ADNG-Net achieves better performance than BaseG, indicating the promotion of the denoiser for the training of the generator.

3) Impact of Latent Input Z: In this work, we assume the latent variable $\mathbf{Z}$ of the generator input to be dependent on the clean HSI $\mathbf{X}$, and thus, we verify its reasonability. Specifically, we train a variant of the ADNG-Net with signal-independent input $\mathbf{Z}$, i.e., white Gaussian noise, denoted as ADNG-Net$_{\mathrm{SI}}$, and compare its performance with the full model. As is shown in Fig. 11(d), the noise generated by the ADNG-Net$_{\mathrm{SI}}$ generator exhibits some fixed patterns and struggles to express the SD noise. Instead, the designed SD input gives rise to a more detailed generated noise with signal dependency, as shown in Fig. 11(b). The corresponding quantitative results are listed in Table V, which shows that the proposed strategy performs better with respect to all the metrics.

V. DISCUSSION OF SEMI-SUPERVISED EXTENSION

In this section, we further discuss the semi-supervised extension of our ADNG-Net. Given the difficulty of acquiring paired data in real-world scenarios, we can further adopt a semi-supervised training scheme into ADNG-Net, leveraging unpaired data. Specifically, we use the following adversarial loss, instead of 4, i.e., the first term in 10, under this semi-supervised setting:

$$\begin{aligned}\min_{R,G}\max_{D_J}\Big\{&\mathcal{L}_{\mathrm{gan}}(R,G,D_J)=2\mathbb{E}_{(\mathbf{X},\mathbf{Y})}\left[D_J(\mathbf{X},\mathbf{Y})\right]\\&-\alpha\mathbb{E}_{(\hat{\mathbf{X}},\mathbf{Y})}\left[D_J(\hat{\mathbf{X}},\mathbf{Y})\right]-(1-\alpha)\mathbb{E}_{(\mathbf{X},\hat{\mathbf{Y}})}\left[D_J(\mathbf{X},\hat{\mathbf{Y}})\right]\\&-\alpha\mathbb{E}_{\mathbf{Y_u}}\left[D_J(R(\mathbf{Y_u}),\mathbf{Y_u})\right]-(1-\alpha)\mathbb{E}_{\mathbf{X_u}}\left[D_J(\mathbf{X_u},G(\mathbf{X_u}))\right]\Big\}\end{aligned}\tag{14}$$

where $(\mathbf{X},\mathbf{Y})$ denotes the paired clean and noisy data, and $\mathbf{X_u}$ and $\mathbf{Y_u}$ denote the clean and noisy HSIs, which are not paired.

We conduct experiments using 1/2 and 1/4 of the total paired samples from the Realistic Dataset [15], each combined with

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a large number of unpaired data. The results are reported in Table VII. It can be seen that even with only 1/4 of the total paired samples (approximately 1430 paired samples and 4300 unpaired samples), the trained generator still has a promising performance. This confirms the strong generalization ability of our framework, even under limited supervision.

VI. Conclusion

In this article, we have proposed, for the first time, a novel DL-based approach for generating realistic HSI noise. Unlike traditional methods relying on predefined coarse hypotheses of noise distribution, our approach learns the noise distribution implicitly from data. In specific, we have proposed to estimate the joint distribution from the perspectives of noise generation and noise removal, leveraging a denoiser to aid the training of the generator. To further enhance generation performance, we have made some novel designs for the generator considering the specific properties of HSI noise, as well as the loss functions. Specifically, we have designed a transformer-based generator with SD random input, which can fully make use of local and global spatial information from the clean HSI, as well as the spectral information. We have also introduced a GAN loss and a KL divergence loss to constrain the statistical characteristics of the generated noise for efficient training. Experiments on different real HSI data have demonstrated the effectiveness of the proposed method, showing that our approach produces HSI noise that is both realistic and diverse, which can facilitate the training of different HSI denoisers and enhance the denoising performance on various HSI datasets.

Though achieved promising results, the proposed HSI noise generation method still has room to improve. On one hand, we will explore more advanced generative models for HSI noise generation, such as diffusion-based models [84], [85], [86]. On the other hand, our framework possesses a certain degree of generality with respect to HSI degradations. It holds promise for being extended to the simulation of other degradation scenarios, such as spectral and spatial low-resolution HSIs, thereby serving as an efficient tool to generate more realistic and diverse training data for related reconstruction tasks [87], [88].

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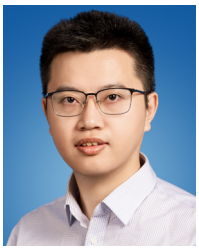
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